

WEEK-4 PRACTICAL

The screenshot displays the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' and 'INSTANCE' sections. The central editor shows a SQL script with four lines: 1. `drop database if exists bookflow_db;`, 2. `CREATE DATABASE bookflow_db;`, 3. `USE bookflow_db;`, and 4. `SELECT DATABASE();`. The 'Result Grid' tab is active, showing the output of the queries. The 'Output' tab is also visible, displaying a table of action results.

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration | **Schemas**

Information

No object selected

Result Grid

#	Time	Action	Message	Duration / Fetch
1	14:37:18	Drop database if exists bookflow_db	Error Code: 1064. You have an error in your SQL synta...	0.000 sec
2	14:37:35	drop database if exists bookflow_db	8 row(s) affected	0.203 sec
3	14:37:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
4	14:37:54	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:38:31	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Output

Action Output

Query Completed

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Week_1_practical SQL File 3* SQL File 4*

Limit to 1000 rows

MANAGEMENT

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Administration Schemas

No object selected

```

13     genre VARCHAR(100),
14
15     description TEXT,
16
17     published_year INT
18 };
19 • describe Books;

```

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result 2 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 2	14:37:35	drop database if exists bookflow_db	8 row(s) affected	0.203 sec
✓ 3	14:37:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
✓ 4	14:37:54	USE bookflow_db	0 row(s) affected	0.000 sec
✓ 5	14:38:31	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 6	14:39:13	CREATE TABLE Books (book_id INT AUTO_INC...	0 row(s) affected	0.047 sec
✓ 7	14:39:55	describe Books	6 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

Query Completed

MySQL Workbench

MySQL Model x

Local instance MySQL80 x

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

Week_1_practicalSQL File 3*SQL File 4* x

SQLAdditions

MANAGEMENT

Server Status

Client Connections

Users and Privileges

Status and System Variables

Data Export

Data Import/Restore

INSTANCE

Startup / Shutdown

Server Logs

AdministrationSchemas

Information

No object selected

112author,

113genre,

114published_year

115FROM Books

116ORDER BY book_id;

117

118

Result Grid

Filter Rows:

Edits

Export/Import:

Result Grid

Form Editor

book_idtitleauthorgenre

1The MartianAndy WeirScience Fiction

2DuneFrank HerbertScience Fiction

3The Hitchhikers Guide to the GalaxyDouglas AdamsScience Fiction

4Pride and PrejudiceJane AustenRomance

5The Great GatsbyF. Scott FitzgeraldClassic

6FoundationIsaac AsimovScience Fiction

7The HobbitJ.R.R. TolkienFantasy

Books 3 x

Apply

Context Help

Snippets

Output

Action Output

#TimeActionMessageDuration / Fetch

414:37:54USE bookflow_db0 row(s) affected0.000 sec

514:38:31SELECT DATABASE() LIMIT 0, 10001 row(s) returned0.000 sec / 0.000 sec

614:39:13CREATE TABLE Books (book_id INT AUTO_INC...0 row(s) affected0.047 sec

714:39:55describe Books6 row(s) returned0.015 sec / 0.000 sec

814:40:52INSERT INTO Books (title, author, genre, ...10 row(s) affected Records: 10 Duplicates: 0 Wami...0.032 sec

914:41:19SELECT book_id, title, author, genre, pu...10 row(s) returned0.000 sec / 0.000 sec

Object Info

Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

MySQL Model xLocal instance MySQL80 x

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

MANAGEMENT

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Administration

Information

Week_1_practicalSQL File 3*SQL File 4*

Limit to 1000 rows

SQLAdditions

113genre,
114published_year
115FROM Books
116ORDER BY book_id;
117ALTER TABLE Books
118ADD COLUMN embedding JSON;
119DESCRIBE Books;
120

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

No object selected

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Result GridForm Editor

Result 4 xRead OnlyContext HelpSnippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	14:41:19	SELECT book_id, title, author, genre, publi...	10 row(s) returned	0.000 sec / 0.000 sec
10	14:43:29	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
11	14:44:04	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec

Object InfoSession

Query Completed

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Week_1_practical SQL File 3* SQL File 4* SQLAdditions

MANAGEMENT

- Server Status
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INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Result Grid Filter Rows: Edit: Export/Import: Result Grid Form Editor

```
149 WHERE book_id = 10;
150 SELECT
151     book_id,
152     title,
153     embedding
154 FROM Books
155 ORDER BY book_id;
156
```

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]

Books 5 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 20	14:50:57	UPDATE Books SET embedding = JSON_ARRAY(0.20...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.015 sec
✓ 21	14:51:08	UPDATE Books SET embedding = JSON_ARRAY(0.70...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.016 sec
✓ 22	14:51:19	SELECT book_id, title, embedding FROM Book...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

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MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Week_1_practical SQL File 3* SQL File 4* SQLAdditions

MANAGEMENT

- Server Status
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INSTANCE

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Administration

No object selected

Information

Result Grid

	book_id	title	author	genre	l2_distance
1	1	The Martian	Andy Weir	Science Fiction	0
8	8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	2	Dune	Frank Herbert	Science Fiction	0.0866

Result 6 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	14:51:08	UPDATE Books SET embedding = JSON_ARRAY(0.70...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.016 sec
22	14:51:19	SELECT book_id, title, embedding FROM Book...	10 row(s) returned	0.000 sec / 0.000 sec
23	14:56:26	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.016 sec / 0.000 sec

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Week_1_practical SQL File 3* SQL File 4* SQLAdditions

MANAGEMENT

- Server Status
- Client Connections
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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Result Grid

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 7 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
22	14:51:19	SELECT book_id, title, embedding FROM Book...	10 row(s) returned	0.000 sec / 0.000 sec
23	14:56:26	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.016 sec / 0.000 sec
24	15:03:59	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

File Edit Selection View Go Run ... Bookflow_API

Explorer

- Bookflow_API
- node_modules
- package-lock.json
- package.json

```
{
  "name": "bookflow-api",
  "version": "1.0.0",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "express": "^5.1.0",
    "mysql2": "^3.14.3"
  }
}
```

Problems Output Debug Console Terminal Ports

PS C:\Users\arike\Bookflow_API> npm install

added 78 packages, and audited 79 packages in 9s

29 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities

PS C:\Users\arike\Bookflow_API>

Build with Agent

AI responses may be inaccurate

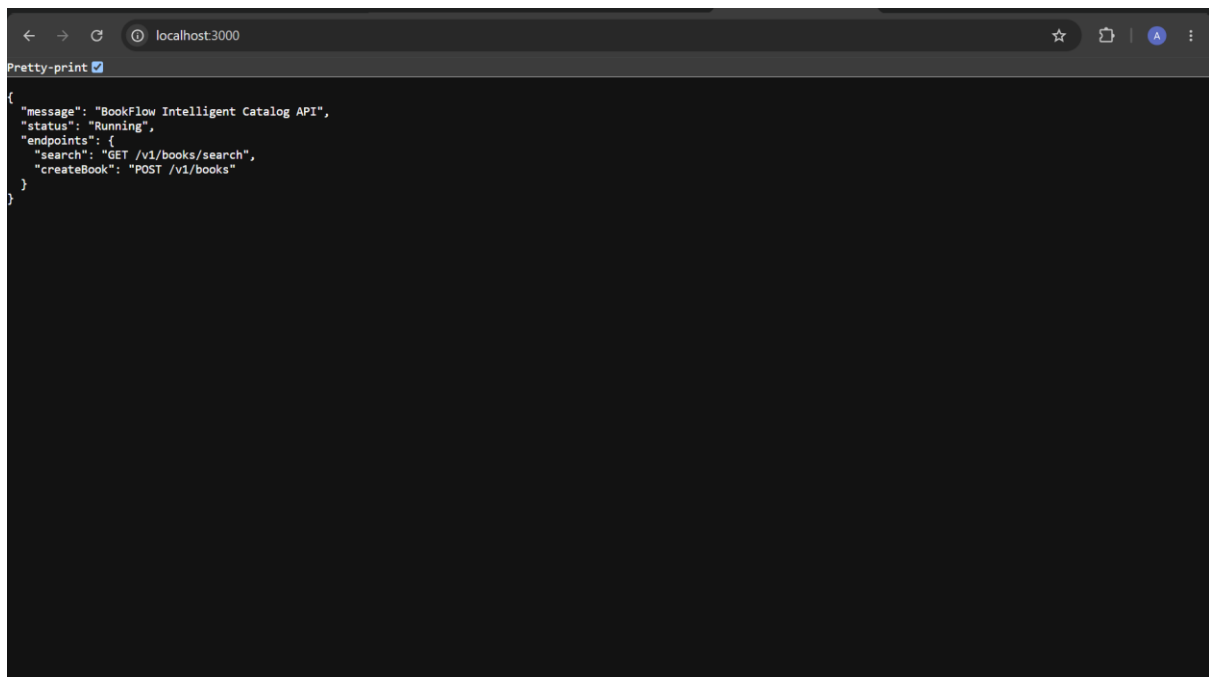
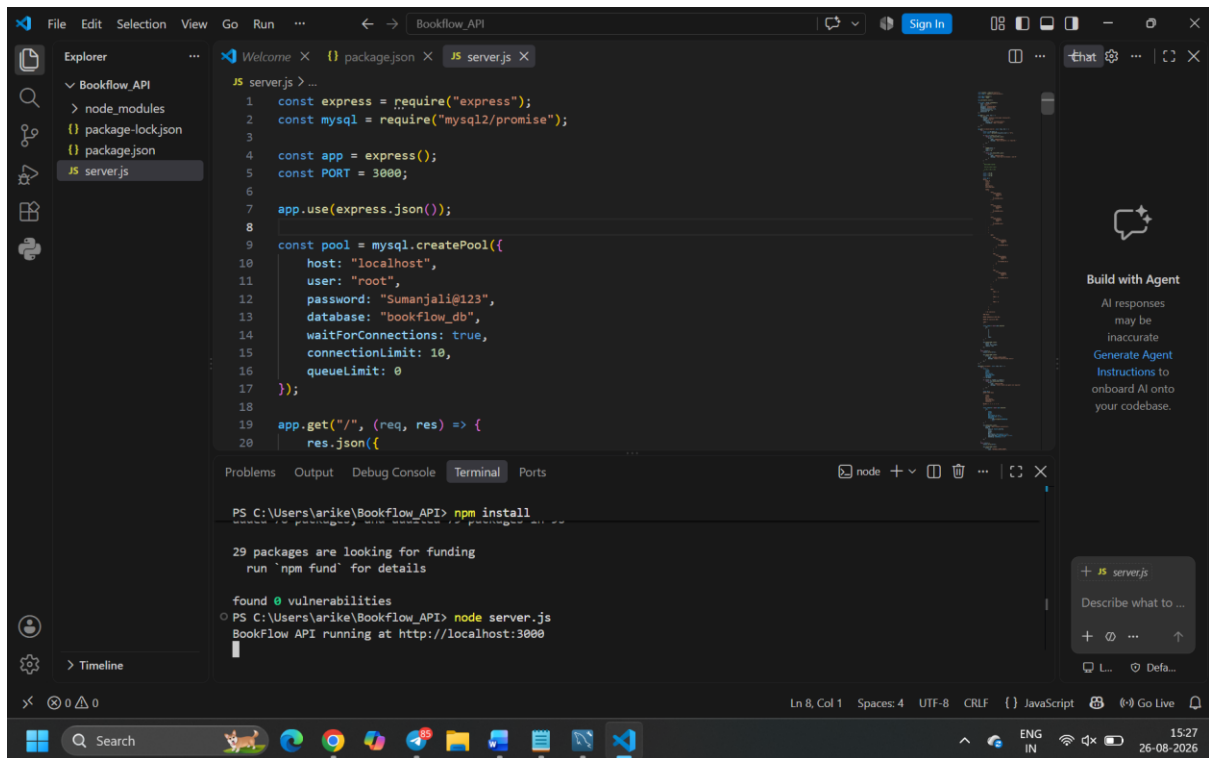
Generate Agent Instructions to onboard AI onto your codebase.

+ package.json

Describe what to ...

+ ...

Go Live



Swagger Editor

Powered by SMARTIDEAS

File

Edit

Generate Server

Generate Client

About

Sign in to Swagger

Try Swagger

1openapi: 3.0.3

2

3info:

4 title: BookFlow Intelligent Catalog API

5 version: 1.0.0

6 description: |

7 REST API for semantic book search.

8

9 The API searches books using mock

10 three-dimensional vectors stored in MySQL.

11

12 servers:

13 - url: http://localhost:3000

14

15 tags:

16 - name: Books

17 - description: Book operations

18

19 paths:

20

21 /v1/books/search:

22 get:

23 tags:

24 - Books

25

26 summary:

27 Search books using semantic similarity

28

29 parameters:

30

31 - name: q

BookFlow Intelligent Catalog API

1.0.0OAS 3.0

REST API for semantic book search.

The API searches books using mock three-dimensional vectors stored in MySQL.

Servers

http://localhost:3000

Books Book operations

GET /v1/books/search Search books using semantic similarity

POST /v1/books Create a new book

Schemas

Curl

curl -X 'GET' \

'http://localhost:3000/v1/books/search?q=Stories%20about%20space&limit=3' \

-H 'accept: application/json'

Request URL

http://localhost:3000/v1/books/search?q=Stories%20about%20space&limit=3

Server response

CodeDetails

200

Response body

{

"description": "An astronaut struggles to survive alone on

"published_year": 2011,

"similarity": 1

},

{

"book_id": 2,

"title": "Dune",

"author": "Frank Herbert",

"genre": "Science Fiction",

"description": "A story involving planets, politics, surviv

al and space.",

"published_year": 1965,

"similarity": 0.9986

},

{

"book_id": 8,

"title": "Enders Game",

"author": "Orson Scott Card",

"genre": "Science Fiction",

"description": "A young student is trained for an interstel

lar conflict.",

Swagger Editor

File Edit Generate Server Generate Client About

Sign in to Swagger Try Swagger

```
1 openapi: 3.0.3
2
3 info:
4   title: BookFlow Intelligent Catalog API
5   version: 1.0.0
6   description: |
7     REST API for semantic book search.
8
9   The API searches books using mock
10  three-dimensional vectors stored in MySQL.
11
12 servers:
13   - url: http://localhost:3000
14
15 tags:
16   - name: Books
17     description: Book operations
18
19 paths:
20   /v1/books/search:
21     get:
22       tags:
23         - Books
24       summary:
25         Search books using semantic similarity
26       parameters:
```

Request URL

http://localhost:3000/v1/books

Server response

Code Details

201

Response body

```
{
  "message": "Book created successfully",
  "book": {
    "book_id": 11,
    "title": "Project Hail Mary",
    "author": "Andy Weir",
    "genre": "Science Fiction",
    "description": "A science fiction survival story involving spac...",
    "published_year": 2021,
    "embedding": [
      0.03,
      0.02,
      0.12
    ]
  }
}
```

Response headers

```
content-length: 253
content-type: application/json; charset=utf-8
```

Responses

Code	Description	Links
201	Book created successfully	No links

MySQL Workbench

MySQL Model Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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INSTANCE

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Administration

Information

No object selected

Week_1_practical SQL File 3* SQL File 4*

Limit to 1000 rows

SQL

```
311 SELECT
312     book_id,
313     title,
314     author,
315     genre,
316     published_year
317 FROM Books
318 WHERE title = 'Project Hail Mary';
319
```

Result Grid

book_id	title	author	genre	published_year
11	Project Hail Mary	Andy Weir	Science Fiction	2021
NULL	NULL	NULL	NULL	NULL

Books 8 x

Apply Revert

Context Help Snippets

Output

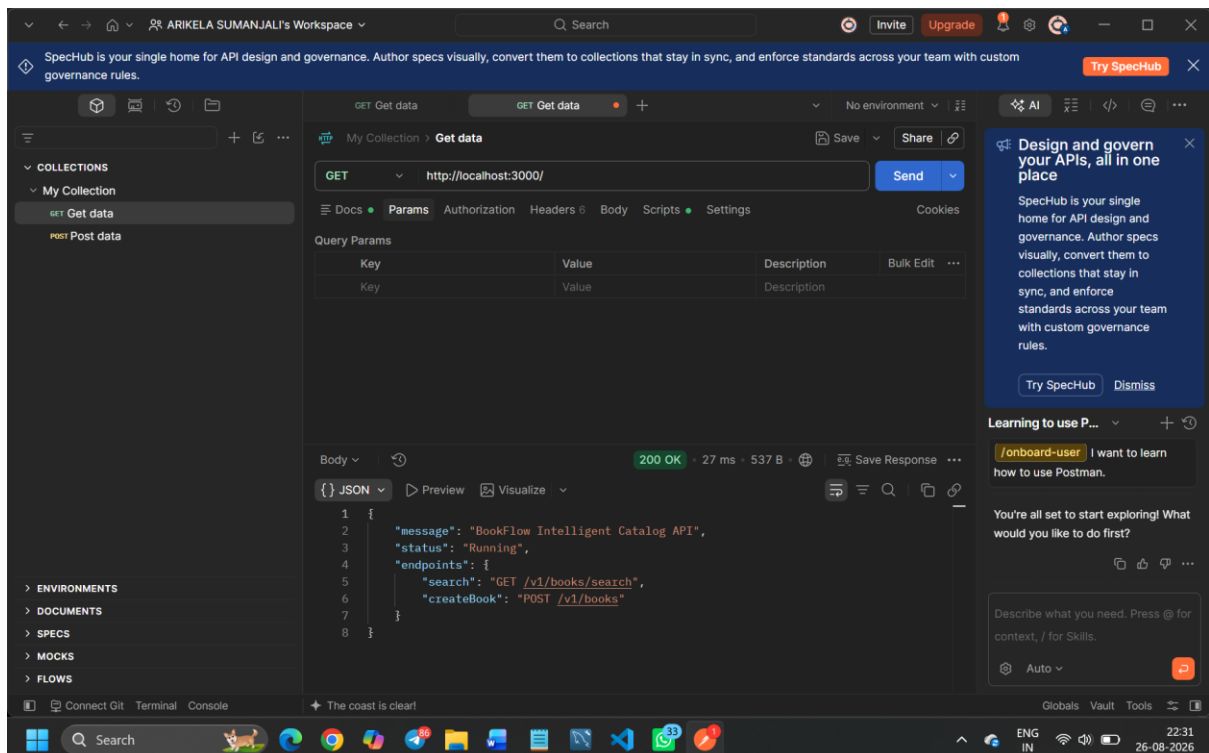
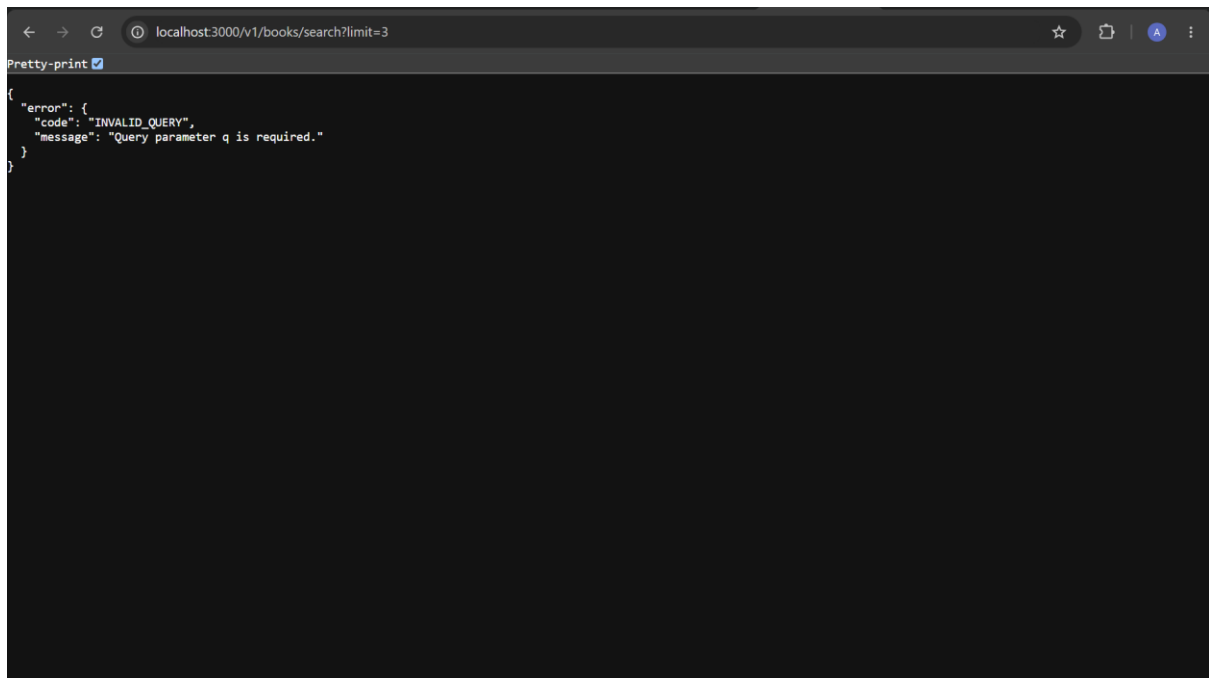
Action Output

#	Time	Action	Message	Duration / Fetch
23	14:56:26	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.016 sec / 0.000 sec
24	15:03:59	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.000 sec / 0.000 sec
25	21:30:47	SELECT book_id, title, author, genre, publi...	1 row(s) returned	0.110 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.



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My Collection > Get data

GET [Send](#)

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
q	Stories%20about%20space		
limit	3		

Body

200 OK · 78 ms · 1.01 KB

```
1 {
2   "query": "Stories about space",
3   "count": 3,
4   "results": [
5     {
6       "book_id": 1,
7       "title": "The Martian",
8       "author": "Andy Weir",
9       "genre": "Science Fiction",
10      "description": "An astronaut struggles to survive alone on Mars.",
11      "published_year": 2011,
12      "similarity": 1
13    },
14    {
15      "book_id": 2
```

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Learning to use P...

/onboard-user I want to learn how to use Postman.

You're all set to start exploring! What would you like to do first?

Describe what you need. Press @ for context, / for Skills.

Auto

Connect Git Terminal Console

The coast is clear!

Globals Vault Tools

22:33 26-08-2026

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My Collection > Get data

GET [Send](#)

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
limit	3		

Body

400 Bad Request · 7 ms · 481 B

```
1 {
2   "error": {
3     "code": "INVALID_QUERY",
4     "message": "Query parameter q is required."
5   }
6 }
```

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Auto

Connect Git Terminal Console

Thinking...

Globals Vault Tools

22:34 26-08-2026

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My Collection > Get data

GET http://localhost:3000/v1/books/search?q=space&limit=100 [Send](#)

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
q	space		
limit	100		

Body

```
1 {
2   "error": {
3     "code": "INVALID_LIMIT",
4     "message": "limit must be between 1 and 50."
5   }
6 }
```

400 Bad Request · 11 ms · 482 B · [Save Response](#)

JSON Preview Debug with AI

ENVIRONMENTS DOCUMENTS SPECS MOCKS FLOWS

Connect Git Terminal Console

Globals Vault Tools

22:36 26-08-2026

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My Collection > Post data

POST http://localhost:3000/v1/books [Send](#)

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "title": "Project Hail Mary",
3   "author": "Andy Weir",
4   "genre": "Science Fiction"
5 }
```

Body

```
2 {
3   "message": "Book created successfully",
4   "book": {
5     "book_id": 12,
6     "title": "Project Hail Mary",
7     "author": "Andy Weir",
8     "genre": "Science Fiction",
9     "description": null,
10    "published_year": null,
11    "embedding": null
12  }
13 }
```

201 Created · 146 ms · 595 B · [Save Response](#)

JSON Preview Visualize

ENVIRONMENTS DOCUMENTS SPECS MOCKS FLOWS

Connect Git Terminal Console

Globals Vault Tools

The coast is clear

22:36 26-08-2026

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My Collection > Post data

POST http://localhost:3000/v1/books

Body

```
1 {
2   "title": "Test Book"
3 }
```

400 Bad Request · 13 ms · 487 B

Body

```
1 {
2   "error": {
3     "code": "INVALID_BOOK",
4     "message": "title, author and genre are required."
5   }
6 }
```

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Auto

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My Collection > Get data

GET http://localhost:3000/v1/books/search?q=Project%20Hail%20Mary&limit=3

Query Params

Key	Value	Description	Bulk Edit
q	Project%20Hail%20Mary		
limit	3		

Body

```
1 {
2   "query": "Project Hail Mary",
3   "count": 3,
4   "results": [
5     {
6       "book_id": 1,
7       "title": "The Martian",
8       "author": "Andy Weir",
9       "genre": "Science Fiction",
10      "description": "An astronaut struggles to survive alone on Mars.",
11      "published_year": 2011,
12    }
13  ]
14 }
```

200 OK · 33 ms · 1.01 KB

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Learning to use P...

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You're all set to start exploring! What would you like to do first?

Describe what you need. Press @ for context, / for Skills.

Auto

Fix status code assertion

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Week_1_practical SQL File 3* SQL File 4* x SQLAdditions

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Limit to 1000 rows

```
320 book_id,  
321 title,  
322 author,  
323 genre,  
324 published_year,  
325 embedding  
326 FROM Books  
327 ORDER BY book_id;  
328
```

Result Grid

book_id	title	author	genre	published_year
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1977
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
11	Project Hail Mary	Andy Weir	Science Fiction	2021
12	Project Hail Mary	Andy Weir	Science Fiction	2021
* NULL	NULL	NULL	NULL	NULL

Books 9 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 24	15:03:59	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 25	21:30:47	SELECT book_id, title, author, genre, publi...	1 row(s) returned	0.110 sec / 0.000 sec
✓ 26	22:53:36	SELECT book_id, title, author, genre, publi...	12 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.